5G RAN

5G LAN Transport Feature Parameter Description

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Contents

1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes

  Changes in functions and their corresponding parameters

- Editorial changes

  Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft A (2025-12-31)

This issue does not introduce any changes to 5G RAN9.1 01 (2025-03-10).

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following feature.

Feature ID	Feature Name	Chapter/Section
FOFD-050209	5G LAN Transport	**4 5G LAN Transport**

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
5G LAN Transport	Supported only in SA networking	**4 5G LAN Transport**

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

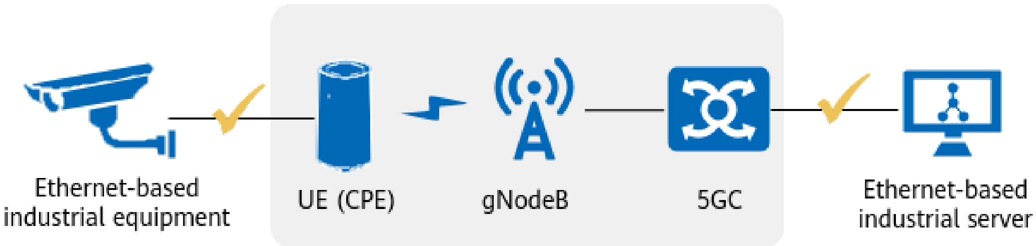
Function Name	Difference	Chapter/Section
5G LAN Transport	None	**4 5G LAN Transport**

3 Overview

5G networks are revolutionizing industrial communications, delivering unparalleled high bandwidth and low latency. Already, wireless 5G connections are replacing wired interconnection in industrial parks. Compared with wired network deployment and maintenance, this significantly reduces costs, enables higher mobility and flexibility, and improves production efficiency. Currently, industrial wired interconnection networks employ traditional IP-based topologies as well as a range of industrial Ethernet-based topologies (such as the PLC control system, PLC indicating programmable logic controller). Frequently-used Ethernet protocols include generic object oriented substation event (GOOSE), EtherCAT, Profinet, and time sensitive networking (TSN).

3GPP Release 15 introduced Ethernet PDU session setup, which supports data transmission over the Ethernet protocol, as shown in **Figure 3-1**. 5G base stations support Ethernet PDU session setup, which means that 5G networks can be applied to industrial Ethernet networking scenarios.

Figure 3-1 Industrial Ethernet-based topology



4 5G LAN Transport

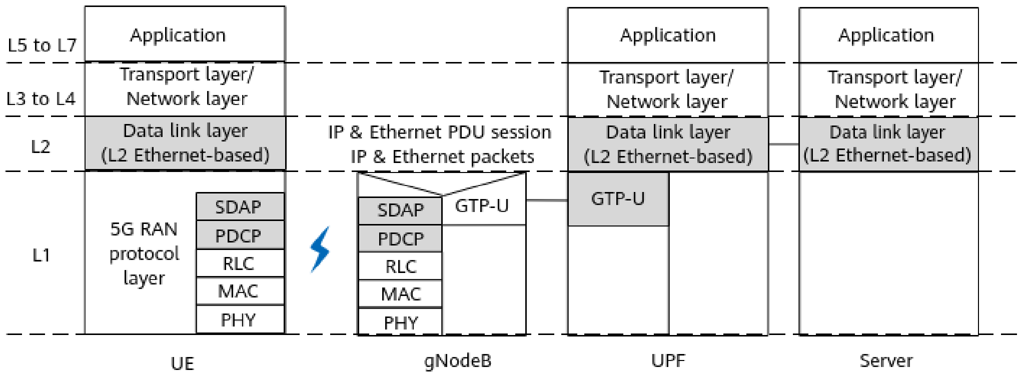
4.1 Principles

This feature is controlled by the **ETHERNET_PDU_SESSION_SW** option of the **NRCellAlgoSwitch**.***ToBusinessServiceSwitch*** parameter.

Ethernet Layer 2 Forwarding

The gNodeB supports Ethernet data packet forwarding and therefore can process industrial protocol applications based on Ethernet Layer 2 forwarding. **Figure 4-1** shows the Ethernet Layer 2 forwarding protocol stack. When the gNodeB forwards Ethernet data packets, the SDAP, PDCP, and GTP-U do not process IP data packets.

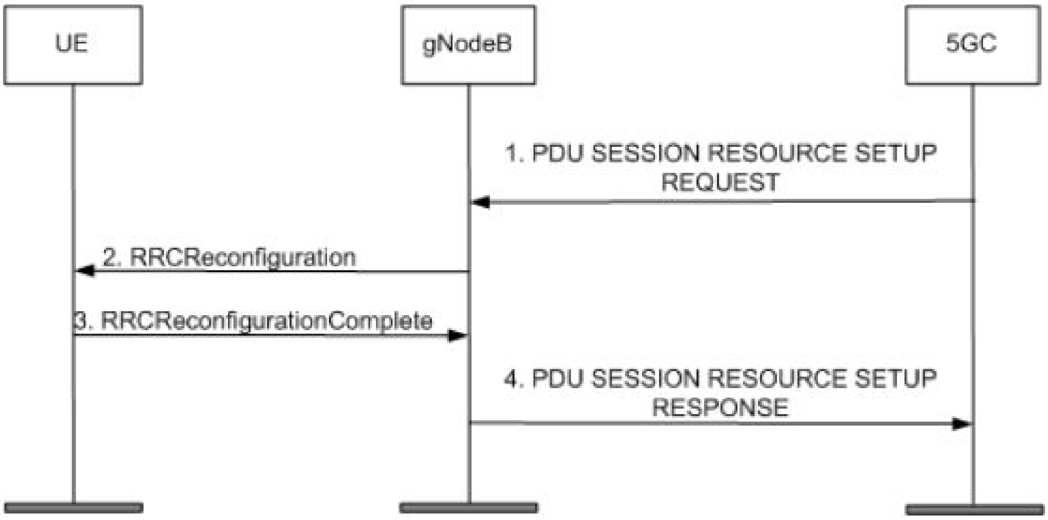
Figure 4-1 Ethernet Layer 2 forwarding protocol stack



Ethernet PDU Session Setup

The gNodeB sets up an Ethernet PDU session based on the PDU SESSION RESOURCE SETUP REQUEST message sent by the 5GC. **Figure 4-2** shows the setup procedure. For details about the PDU session, see section 8.2.1 "Session Resource Setup" in 3GPP TS 38.413 V15.5.0.

Figure 4-2 Ethernet PDU session setup procedure



1. The 5GC sends a PDU SESSION RESOURCE SETUP REQUEST message to the gNodeB.
2. The gNodeB sets up a data radio bearer (DRB) based on QoS flow information carried in the PDU SESSION RESOURCE SETUP REQUEST message, and then sends an RRCReconfiguration message to the UE. In the case of an Ethernet PDU session, the PDU SESSION TYPE IE carried in the PDU SESSION RESOURCE SETUP REQUEST message indicates Ethernet. In this case, IP-based features do not take effect on the PDU session.
3. After setting up the DRB, the UE sends an RRCReconfigurationComplete message to the gNodeB.
4. The gNodeB sends the 5GC a PDU SESSION RESOURCE SETUP RESPONSE message indicating that the PDU session setup is complete.

NOTE

NSA UEs that have accessed the network do not support Ethernet PDU session setup.

4.2 Network Analysis

4.2.1 Benefits

5G is available to replace wired networking for an industrial Ethernet-based network.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

This function has no impact on the network.

4.3 Requirements

4.3.1 Licenses

The following are license requirements for 5G LAN Transport.

Feature ID	Feature Name	Model	Sales Unit
FOFD-050209	5G LAN Transport	NR0S005LAN00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
TDD	Uplink preallocation for sparse-packet services	• **SPARSE_DIFF_PROC_SW** option of the **gNodeBParam**.***ServiceDiffProcSwitch*** parameter • **SPARSE_DIFF_SCH_SW** option of the **NRDUCellAlgoSwitch**.***ServiceDiffSwitch*** parameter	*Uplink Scheduling*	This function does not take effect if an Ethernet PDU session has been set up for the UE.
TDD	Basic VoNR functions	**VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter	*VoNR*	This function does not take effect if an Ethernet PDU session has been set up for the UE.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations. 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Others

The UE and 5GC support Ethernet packet transmission.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Parameter Name	Parameter ID	Setting Notes
To Business Service Switch	**NRCellAlgoSwitch**.***ToBusinessServiceSwitch***	Select the **ETHERNET_PDU_SESSION_SW** option.

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Activation Command Examples

```
MOD NRCELLALGOSWITCH: NrCellId=7, ToBusinessServiceSwitch=ETHERNET_PDU_SESSION_SW-1;
```

Deactivation Command Examples

```
MOD NRCELLALGOSWITCH: NrCellId=7, ToBusinessServiceSwitch=ETHERNET_PDU_SESSION_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

After activating the feature, perform the following steps to check whether this feature has taken effect:

1. Start an NG interface tracing task by choosing **NG Interface Trace** on the MAE-Access.
2. Check whether the *PDU Session Type* IE in the PDU SESSION RESOURCE SETUP REQUEST message sent from the 5GC to the gNodeB indicates an Ethernet session type. If this is true and the gNodeB sends a PDU SESSION RESOURCE SETUP RESPONSE message to the 5GC, the Ethernet PDU session has been successfully set up for the UE.

4.4.3 Network Monitoring

None

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- *Uplink Scheduling*
- *VoNR*
- *License Control Item Lists*
- 3GPP TS 38.413 V15.5.0: "Session Resource Setup"
- 3GPP TS 38.104 V17.8.0: "General"